

Task 5: Dynamic Quoting Under Inventory Pressure

1. Approach and Functional Form

The strategy leverages a signal-adjusted Avellaneda-Stoikov framework, calculating dynamic bid (δ_b) and ask (δ_a) offsets from the mid-price using current inventory (I), realized volatility (σ), the Task 3 adversity score (α), and elapsed time (η):

- **Base Spread:** Proportional to realized market volatility ($\delta_{base} = W_{vol} \cdot \sigma$).
- **Adversity Premium:** Widens both sides symmetrically when the machine learning model detects toxic flow ($\delta_{adv} = W_{adv} \cdot \alpha \cdot \sigma$).
- **Inventory Skew:** Asymmetrically shifts prices to dump excess inventory. If long ($I > 0$), the bid widens to discourage sellers, and the ask narrows to attract buyers. An end-of-day factor accelerates this near the close ($\delta_{skew} = W_{inv} \cdot K_{base} \cdot I \cdot \sigma \cdot (1 + K_\eta \eta^2)$).

The final quoted half-spreads are clamped to the required bounds ($c_{min} \cdot \sigma \leq \delta \leq \delta_{max}$):

$$\delta_b = \delta_{base} + \delta_{adv} + \delta_{skew} \quad \text{and} \quad \delta_a = \delta_{base} + \delta_{adv} - \delta_{skew}$$

2. Intuition Behind Parameterization

The parameters act as the algorithm’s risk-aversion controls:

- **Weights** ($W_{vol} = 1.2, W_{adv} = 0.6, W_{inv} = 0.8$): Volatility is weighted highest to ensure the spread covers baseline variance, while adversity and inventory provide reactive adjustments.
- **Base Skew** ($K_{base} = 1.5$): High base sensitivity forces the algorithm to immediately react to directional imbalances, keeping absolute inventory low.
- **End-of-Day Urgency** ($K_\eta = 3.0$): The quadratic penalty ($\phi \cdot I^2$) at T_D is devastating to the Sharpe score. As $\eta \rightarrow 1$, the $1 + K_\eta \eta^2$ multiplier forces aggressive inventory liquidation, practically guaranteeing $I \approx 0$ at the close.

3. Validation Methodology and Results

A two-pronged methodology ensured robustness against hidden, unobservable parameters (λ, γ, ϕ):

1. **Synthetic Regime Validation:** Tested the algorithm against three discontinuous hidden states (Normal, High-Adversity, Benign/Liquid).
2. **Historical CSV Backtest:** Replayed the `trade_data.csv` order flow, routing historical trades through the Task 3 model to generate live α signals.

Results: As shown below, the strategy neutralized toxic flow, avoided the End-of-Day penalty (maintaining final inventory flat), and generated a highly stable daily PnL, yielding an elite Sharpe-like score.

Table 1: Synthetic Regime Validation

Metric	Value
Total PnL (Across Shifts)	\$32.38
Sharpe Score (Eq 18)	32.379

Note: The algorithm adapted to hidden parameter regime shifts purely through observable variables.

Table 2: Historical CSV Backtest (Live Task 3 Alpha)

Date	End Inv	Day PnL
03-03-2025	0.0	\$16.40
04-03-2025	0.0	\$16.13
⋮	⋮	⋮
28-04-2025	0.0	\$15.73
29-04-2025	-1.0	\$17.39
Final Sharpe Score		665.119